

Indoor Mesh Network Wi-fi Trilateration Location Tracking

Spring 2025 ECE 56800 Project Final Report

Wi-Fi Compass

Group 14

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ABSTRACT

The ability to determine the precise location of indoor localization remains a significant challenge due to signal attenuation and multipath interference, rendering traditional GPS ineffective and alternative high-precision solutions like Ultra-wideband (UWB) cost-prohibitive. This project presents an affordable, low-power indoor positioning system leveraging the IEEE 802.11mc Fine Time Measurement (FTM) protocol on commodity Wi-Fi hardware. By measuring the Round-Trip Time (RTT) of Wi-Fi signals between a mobile initiator and three stationary ESP32-S3 anchor nodes, the system calculates relative distances and employs a custom trilateration algorithm to estimate a 2D coordinate position. To mitigate indoor multipath noise, the data is processed using boundary rejection alongside median and exponential moving average (EMA) filtering. Real-world testing in a 60-square-meter obstacle-dense classroom demonstrated successful real-time tracking via a localized web server, achieving an average localization error of 1.33 meters. These results validate the viability of FTM-based Wi-Fi mesh networks for accessible, high-precision indoor tracking.

1 INTRODUCTION

Accurate indoor localization is increasingly vital for applications ranging from asset tracking in warehouses to personal navigation in crowded venues. While outdoor localization is largely solved by GPS, indoor environments present unique challenges. Structural obstacles, multipath interference, and signal attenuation often render traditional GPS ineffective and disrupt cellular reliability. When a person's line of sight is obstructed by walls, equipment, or crowds, the ability to efficiently determine the location of other entities is drastically reduced.

Although alternative localization methods exist, they often present a trade-off between cost and accuracy. Technologies like Ultra-wideband (UWB) offer high precision but require the deployment of expensive, proprietary hardware tags and anchors. Conversely, standard Wi-Fi Received Signal Strength Indicator (RSSI) methods utilize existing infrastructure but suffer from severe accuracy limitations due to environmental noise and signal degradation.

To bridge this gap, this project leverages the IEEE 802.11mc Fine Time Measurement (FTM) protocol to achieve precise indoor localization using low-power, commodity embedded systems. By utilizing a network of ESP32-S3 microcontrollers acting as stationary anchor nodes, we implemented a system that calculates a mobile device's position through Round-Trip Time (RTT) trilateration. This approach eliminates the need for expensive proprietary hardware

while providing robust, meter-level accuracy, effectively demonstrating how standard Wi-Fi infrastructure can be repurposed into a highly reliable indoor positioning system.

2 BACKGROUND

GPS is commonly used outdoors. By measuring the communication delay between a device and multiple satellites in orbit, localization can be achieved with sub 5-meter accuracy[1]. However, this means that structures like walls, roofs, etc., often interfere with GPS signals, resulting in low accuracy in indoor settings. Prior efforts at improving indoor localization have been made with ultrasound, infrared, Bluetooth, Ultra-wideband (UWB), and Wi-Fi[2]. Ultrasound is vulnerable to multipath-effect where a signal travels multiple paths before it reaches the receiver, which will result in data error. Infrared is vulnerable to light while Bluetooth is easily affected by various barriers.[5] UWB is the protocol that's often used for indoor localization due to its high accuracy. However, it requires additional equipment (tag and anchor) and is more expensive than using the existing Wi-Fi infrastructure. Previously, the Wi-Fi protocol has supported indoor localization through RSSI-based triangulation and fingerprinting methods[3]. However, these methods suffer some of the same issues which affect ultrasound and Bluetooth, where multipath interference and physical objects can reduce the accuracy of the estimation. Some approaches attempt to use machine learning to attempt to account for various sources of interference[3].

3 PROBLEM STATEMENT

Locating other people and devices in crowded areas can be a massive challenge and indoor localization has played a significant role in fields like monitoring, surveillance, navigation, and security[4]. One application is to track patients in a hospital for various reasons like their safety and quality of care[4]. Another application is asset tracking in hospitals, malls, at work, or even at home[4]. However, unlike GPS tracking, indoor localization suffers from numerous obstacles, noisy signals, and moving elements, all of which exacerbate multipath distortion. Existing methods force a tradeoff between precision and cost. As seen in Table 1, standard Bluetooth is ubiquitous but has low localization accuracy. On the other hand, Ultra-Wideband (UWB) offers high precision but requires proprietary, expensive tags.

The IEEE 802.11mc Fine Time Measurement (FTM) protocol bridges this gap by achieving UWB-like precision using standard Wi-Fi hardware. However, implementing FTM on a highly constrained embedded system is a challenge in itself. For example, the ESP32-S3 microcontrollers channels only support Wi-Fi at 2.4 GHz which would cause highly noisy data. Therefore, the goal of our

project is to implement FTM on a commodity microcontroller and overcome the hardware limitation that follows it by using a filtering and trilateration so that the system can extract accurate locations using existing Wi-Fi infrastructure.

	Max. Range	Advantage	Disadvantage
IEEE 802.11 (Wi-Fi 4)	35 m indoor	Widely available, accurate, doesn't require extra hardware	Prone to noise, requires extra processing algorithms
UWB	10-20 m	Immune to interference, high accuracy	Shorter range, extra hardware, high cost
Ultrasound	Couple-tens of meters	Less absorption	Dependence on sensor placement
Bluetooth	100 m	High throughput, reception range, low power consumption	Low localization accuracy, prone to noise

Table 1: Comparison of Indoor Localization Technologies[7]

4 PROPOSED APPROACH

4.1 Fine-Time Measurement

To achieve precise tracking of a mobile microcontroller, this system implemented fine-time measurement (FTM). The IEEE 802.11mc revision introduced FTM to the Wi-Fi standard. This protocol can enable accurate positioning by measuring the round-trip time (RTT) of the Wi-Fi signal between an access point (AP) and an initiator. The propagation speed of the Wi-Fi signal is assumed to be equal to the speed of light. However, the initiator and the AP are not synchronized at granularity of nanoseconds. This would result in a time difference if only a one-way signal was measured. Therefore, FTM uses RTT so that both the AP and the initiator can use their internal clock to measure the elapsed time since they transmitted a frame and received a response. The specific process is described as follows: The initiator first sends an FTM request to the AP. The AP responds with a FTM frame at time t_1 which is received by the initiator at time t_2 . Once the initiator receives the frame, it sends a response to the AP at t_3 and this response is received by the AP at t_4 . To conclude the packet transaction, the AP will send the t_1 and t_4 to the initiator. The initiator will use all four time stamps to calculate the RTT as follows:

$$RTT = (t_4 - t_1) - (t_3 - t_2)$$

As a result, the distance can be calculated as:

$$d = \frac{RTT}{2} \times c$$

Where c is the speed of light[6]. In practice, these RTT measurements contain many outliers that obscure the relevant distance measurements. To reduce the effects of these outliers on our final localization calculation, we first discard any physically impossible

distance measurements that exceed the known bounds of the room. Then we median filter the remaining distance measurements to remove further outliers. While these filters are effective at reducing undesired outliers, they increase the time it takes to achieve valid and stable coordinate estimates.

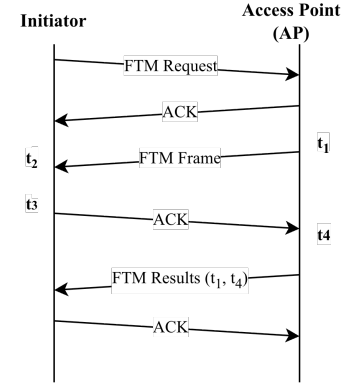


Figure 1: FTM Protocol

4.2 Trilateration

Using the Wi-Fi FTM protocol, our system utilized three ESP32-S3 devices as stationary anchor points and a fourth mobile ESP32-S3 as the initiator. The mobile device transmitted FTM requests to all anchor points as described previously. This allowed the system to measure the distance between the initiator and all three anchor points. This data was then used to continuously calculate the estimated (x,y) position of the initiator in a 2D-plane.

To calculate the location of the point, we begin by defining the coordinates of the three stationary nodes as (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , and the distances from each of the nodes as d_1, d_2, d_3 . Since all of these are assumed to be known, the desired coordinate is the intersection of the three circles of radius d formed around each of the nodes. This forms the following set of equations:

$$(x - x_1)^2 + (y - y_1)^2 = d_1^2$$

$$(x - x_2)^2 + (y - y_2)^2 = d_2^2$$

$$(x - x_3)^2 + (y - y_3)^2 = d_3^2$$

When linearized, these equations become:

$$2x(x_1 - x_2) + 2y(y_1 - y_2) = d_2^2 - d_1^2 - x_2^2 + x_1^2 - y_2^2 + y_1^2$$

$$2x(x_1 - x_3) + 2y(y_1 - y_3) = d_3^2 - d_1^2 - x_3^2 + x_1^2 - y_3^2 + y_1^2$$

That are then converted into the following linear equation:

$$\begin{bmatrix} 2(x_1 - x_2) & 2(y_1 - y_2) \\ 2(x_1 - x_3) & 2(y_1 - y_3) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} d_2^2 - d_1^2 - x_2^2 + x_1^2 - y_2^2 + y_1^2 \\ d_3^2 - d_1^2 - x_3^2 + x_1^2 - y_3^2 + y_1^2 \end{bmatrix}$$

x and y can be solved using the known positions of the anchor points and is reduced to the following:

$$x = \frac{r_3^2 - r_2^2 + X_{max}^2}{2 \cdot X_{max}}$$

$$y = \frac{r_2^2 - r_1^2 + Y_{max}^2}{2 \cdot Y_{max}}$$

Where X_{max} and Y_{max} are reference distances for the distance between two of the anchors. x and y are relative positions to a set point $(0, 0)$. In real applications with interference and varying RTTs, the distances will not trilaterate to an exact point; it may generate a Reuleaux triangle (roughly circular triangle) region that contains the exact location. From this, the algorithm will take the average X and average Y points. In more severe cases, the position may be calculated that it is not in the range of one of the anchors.

While there are methods that exist which could potentially provide for more accurate localization, they come with tradeoffs compared to our implementation. One example of this is a non-linear least squares (NLLS) approach: while our current method relies on the assumption that our distance measurements from all three anchors converge near a single point, NLLS takes into account the fact that each distance measurement contains an error. NLLS then computes an estimate for location by minimizing the error across all anchor nodes. While this has the potential to outperform our implementation, it requires additional computations, requiring more power and CPU time for localization. For our project, we determined that the linear technique provided for sufficient levels of accuracy without compromising on speed or power savings.

4.3 System Implementation

This project was conducted in a 6 by 10-meter (20 by 32-foot) classroom with multiple obstacles (chair, tables, whiteboard, etc.). The three anchors were placed in the room as seen in Figure 2. One of the anchors was chosen to represent the origin. The initiator was moved to pre-measured points in the room. At every point, the initiator would measure its distance from all three anchors, perform trilateration to determine its (x,y) coordinate, and send it to the web server to be displayed.

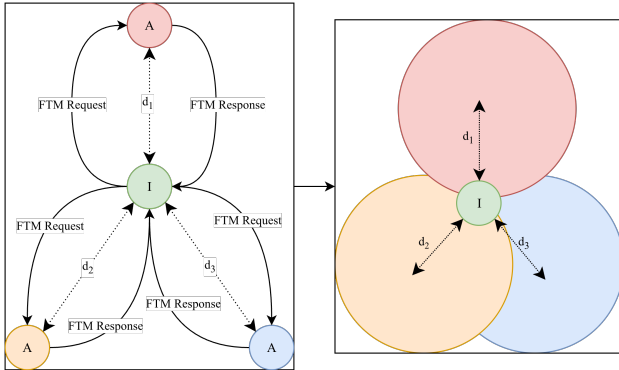


Figure 2: System Diagram

4.3.1 Anchor Nodes. The FTM protocol requires two types of devices: initiators and responders. In our implementation, our anchor nodes act as the responders in the system. An anchor node can be any device that supports the FTM protocol. Here, we utilize 3 ESP32-S3s, which are capable of broadcasting FTM signals over the 2.4ghz WiFi band. Our specific ESP32s provided support for an external antenna. For each of our anchor nodes, we attached dipole antennas to improve the signal strength and quality. The FTM signal emitted from each anchor node is discoverable by our

initiator. When an anchor receives a request frame, it will immediately return a response packet. Because the initiator node processes all the data in our implementation, each anchor can operate in a low-power mode, powering only the components required for WiFi, while disabling other peripherals.

```

1 void main(void){
2     INITIALIZE Non-Volatile Storage (NVS)
3     INITIALIZE TCP/IP network interface
4     INITIALIZE system event loop
5
6     DEFINE Wi-Fi Access Point (AP) parameters
7     SET Wi-Fi operation mode to AP
8     APPLY AP parameters to the Wi-Fi driver
9     START the Wi-Fi hardware driver
10 }

```

4.3.2 Initiator Node. The initiator node in our implementation acts as the mobile node in the system. While we utilized a fourth ESP32-S3 to act as this node, the initiator node can be any device that supports the WiFi Fine Time Measurement Protocol. To measure the distance from each anchor node, the initiator runs on a set of pre-defined SSIDs. If all SSIDs are successfully detected, the initiator then begins the main code loop: For each anchor, the initiator node sends a request packet. After receiving a response from an individual anchor, the initiator then computes the estimate for the distance between the two nodes. After repeating this process for each anchor, the initiator uses each calculated distance to estimate its position in a given area.

To calculate the position of the initiator, we utilize the trilateration approach discussed in the proposed approach section. However, to increase the accuracy of the measurements, our implementation required the introduction of offsets to the distances calculated from each anchor. This is due to the processing delay which is internal to the MAC layer of each anchor node's Wi-Fi stack. As every piece of silicon is unique, the resultant delay can differ between devices, resulting in incorrect, yet consistent measurements from each device. To counteract this, our team experimentally determined the error inherent to each anchor: by placing our initiator at a known distance from each anchor, we were able to measure the resulting error. We were then able to use these values to create offsets for each anchor. We then remeasured the distance reported from each anchor node to confirm the accuracy of our offsets.

```

1 static void ftm_event_handler(Event Data){
2     MATCH responding MAC address to anchor
3
4     if ping successful
5         RawDist = Round Trip Time * 0.14989
6         CalibratedDist = RawDist + offset
7         if CalibratedDist < 0: Set to 0
8
9         Window_Buffer += CalibratedDist
10        Sort buffer
11        MedianDist = Middle value in the buffer
12        Dist=(EMA*MedianDist)+((1-EMA)*PrevDist)

```

```

13     SAVE Dist to the Anchor's profile
14 else
15     PRINT warning ping failed
16     SIGNAL Main Loop to wake up and continue
17 }
18 void main(){
19     INITIALIZE System Storage and Wi-Fi
20
21     START Wi-Fi Scan
22     FIND "anchor1", "anchor2", and "anchor3"
23     SAVE their MAC addresses and Wi-Fi channels
24
25     if any anchor is missing STOP
26
27     while (1){
28         for(anchor){
29             SET Wi-Fi channel to match Anchor
30             SEND FTM Ping request to Anchor
31             WAIT up to 400ms for ftm event
32         }
33
34         r1 = Anchor 1 Dist
35         r2 = Anchor 2 Dist
36         r3 = Anchor 3 Dist
37         X_Pos = ((r3 ^ 2) - (r2 ^ 2) + (X_MAX ^ 2)) / (2 * X_MAX)
38         Y_Pos = ((r2 ^ 2) - (r1 ^ 2) + (Y_MAX ^ 2)) / (2 * Y_MAX)
39         if difference > 0.5m: cap step to +0.5m
40         if difference < -0.5m: cap step to -0.5m
41         Target X = Current X + Clamped X Step
42         Target Y = Current Y + Clamped Y Step
43         Current X = Apply EMA to Target X
44         Current Y = Apply EMA to Target Y
45         Clamp Current X between 0 and Room Width
46         Clamp Current Y between 0 and Room Length
47
48         WAIT 100ms
49     }
50 }

```

4.3.3 Server. To help visualize the real-time location, we created a server on the initiator node. This server takes the calculated distances, as well as the calculated x,y coordinates, and displays them, creating the following UI:

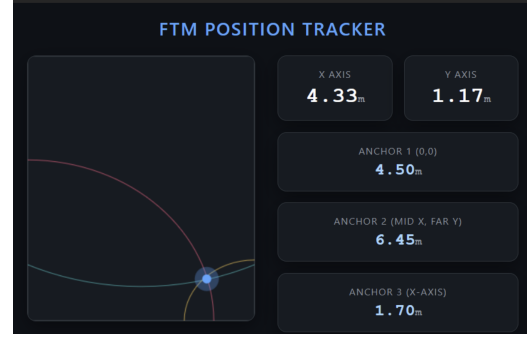


Figure 3: Server Interface

As can be observed in the figure above, the server not only displays the coordinates and distances, but it also creates a Cartesian plane to visually display the radii created by the trilateration. The blue dot indicates the position of the initiator node relative to the origin, which is the bottom-left corner of the Cartesian plane. The location of the blue dot is the centroid of the triangle created by the three intersection points of the circles.

5 EVALUATION

Actual Location	Reported Location	Error (Euclidean)
(3.32m,3.65m)	(1.23m,2.71m)	2.29m
(0.81m,1.78m)	(0m,1.63m)	0.82m
(2.31m,2.13m)	(1.66m ,1.63m)	0.82m
(0m,0m)	(0.35m,1.99m)	2.02m
(2.7m, 4.5m)	(2.69m, 3.82m)	0.68m

Table 2: Measurement Comparison

Avg. Error	Std. Deviation
1.33m	0.76m

Table 3: Measurement Error and Standard Deviation

The evaluation and testing methodology was to select a point, measure the exact distance, and then bring the initiator to the testing location. The initiator would then be allowed to settle for one minute to a more precise location. This was conducted in a medium-sized classroom within a 4-meter by 5-meter area. Although, proportionally the error is large, for larger settings and real applications, the error increases relatively slowly compared to the total size of the room, according to preliminary testing. Testing was also performed to evaluate the best parameters for window size and exponential moving average (EMA) values, which were found to be approximately a 30-sample window and high values for EMA, >0.85. The standard deviation of the positions of 0.76m falls well within the initial limit of 4 meter accuracy. As seen in the above table the position of the initiator node should be easily identified using the project system.

While the average error observed across testing coordinates was 1.33 meters, the error was not uniformly distributed between all points in the environment. For example, the coordinate (3.32m, 3.65m) resulted in a disproportionately high error compared with our other coordinates. This can be attributed to the phenomenon

known as Geometric Dilution of Precision (GDOP). When operating near the bounds of the testing environment, the intersection of each anchor node's distance occurs at much shallower angles. Consequently, any minor fluctuation in the distance measured by a node will result in a massive shift in the estimated position. As a result, fluctuations from sources such as multipath distortion result in a much greater error for measurements near the edges of our testing environment.

6 DISCUSSION

The results demonstrate that accurate localization using WiFi RTT is possible and can be accurate to 4 meters of error. The strength of the design is the simplicity of the embedded system. In an application setting, all the anchor nodes would be WiFi routers; hence, there would be very little need for extra hardware. Additionally, our accuracy could be improved in a real application setting for multiple reasons. First, modern WiFi routers utilize the 5GHz WiFi band in addition to the 2.4GHz WiFi. The ESP32-S3 used in our setup only operates on the 2.4GHz band. Due to this, the bandwidth available to modern routers is much larger than the narrow 40 MHz band available to the ESP32-S3. An increased bandwidth would allow for much faster sampling of distances, which would not only reduce the number of outliers in the distance samples, but it would also allow for more aggressive filtering to remove any remaining outliers. Modern routers also have a much higher signal strength, which would increase the range and also act to decrease the number of physically impossible or outlier distant measurements. These factors would act to reduce the error seen in our setup, while also decreasing the time it takes for the localization calculation to converge to the real world location. The antenna configuration used in this setup is not ideal. Initially, all nodes used the on-board PCB antenna, but we noticed that the orientation of the PCB antenna had a large impact on the quality of distance measurements and often only two of the stationary nodes would produce valid measurements at a time, which greatly impacted our results. To address this we switched to using dipole antennas for each of the stationary nodes, but unfortunately, the moving node used did not have the capability to use an external antenna. The addition of dipole antennas for the stationary nodes improved the consistency of the distance measurements, but the moving node's orientation still had a large impact on the final localization calculation. This issue would be resolved in a real-world setting using modern routers and mobile devices that have much better antennas along with much larger energy outputs. Generally, large public spaces and gatherings would also have more WiFi routers to act as stationary nodes, which would improve the accuracy of the position.

7 POTENTIAL IMPACTS

The device has many potential impacts and applications in the realm of indoor and even outdoor localization, especially given its effective use of low-power micro-controllers. This includes consumer applications, such as tracking friends in concerts, to enterprise applications, such as tracking employee movement productivity, or even tracking athletes on a large court or field. The use case is not limited to just humans; this technology can also be used to

track non-human assets, such as livestock, robots, or packages in a warehouse.

Additionally, our project offers a solid foundation for further development in low-power localization technologies. Our research on FTM tracking can be further improved upon to enable more precise tracking over longer distances.

8 CHANGE OF PLAN

Originally, the plan for the project was to use a computer connected to our mobile initiator node to print out distance measurements from each anchor, and the absolute position in a room. However, to increase ease of use, the mobile node was instead programmed to act as a web server, where a user can now connect to the esp via Wi-Fi and view these values. Additionally, the web server included a representation of the room with each anchor's distance readings overlaid, as well as the final estimated location of the mobile node.

While testing our implementation, we also experimented with adding dipole antennas to our ESP32s. Without the antennas, our team ran into significant issues due to multipath distortion and signal degradation. As three of our four ESP32s had the ability to mount an external antenna, we experimented with using these as our anchors. Compared with the PCB antenna embedded within the ESP32, an external dipole provided much greater signal strength, while also broadcasting more uniformly in every direction.

9 EMBEDDED SYSTEMS ASPECTS

To implement the localization methods utilized by our team, the requirements are that a device supports the WiFi 802.11mc revision or later, where the Wi-Fi Round Trip Time protocols were first implemented. Many embedded devices, such as the ESP32-S3 used by our team, natively support localization. Given the proliferation of Wi-Fi enabled IoT devices in recent times, it is possible, and very easy, for common existing hardware to be repurposed to serve as anchors for localization.

The system also puts emphasis on low-power operation, as the idea is to constantly track a given asset for a continuous duration of time. This is especially true for the initiator node, which is assumed to be in constant movement, and thus powered by a battery instead of a stationary power supply. Given such considerations, the system is limited on power consumption. This project has ample room for optimization in power consumption while maintaining reliable precision and accuracy. In terms of hardware improvements, the latest generation of low-power Wi-Fi SoCs should prove significantly more power-friendly compared to the commonly available ESP32-S3s used for the prototype. As for software, implementing an RTOS that prioritizes fast return-to-sleep times while balancing wireless performance can be utilized to optimize the system for low power operation.

10 TASK BREAKDOWN

In the following table, we describe the contributions of each team member:

Alec Andrulis	Base Code and Calibration
Shubham Saluja	Server and Server Code integration
Jason Shepherd	Testing and Tuning
Eric Min	Hardware procurement
Cecilie Zhang	Documentation
Michael Lee	Testing and Tuning

Table 4: Work Distribution

11 RELATION TO OTHER PROJECT

Not Applicable

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